

Leaf Vein Quantification and Measurements

叶脉定量与测量

Sentence by sentence bilingual version | 逐句中英文对照版

No.	English original	中文翻译
1	EN: For leaf vein imaging, expanded flag leaves were collected from the glasshouse before and after heat treatments and fixed in Formalin Acid Alcohol (FAA) solution (composed of 37% aqueous formaldehyde solution, 70% ethanol, and 5% glacial acetic acid, pH 7.0) for later use.	中文: 为进行叶脉成像，在热处理前后从温室中采集完全展开的旗叶，并将其固定于福尔马林 乙酸 酒精固定液（FAA）中备用。该固定液由 37% 甲醛水溶液、70% 乙醇和 5% 冰乙酸组成，pH 为 7.0。
2	EN: Leaf samples were prepared for imaging following the clearing and staining protocol (Naznin et al., 2026; Scoffoni & Sack, 2013).	中文: 叶片样品按照透明化和染色流程进行成像前处理，方法参考 Naznin et al. (2026) 以及 Scoffoni and Sack (2013)。
3	EN: The widest part of each leaf was excised in 1 cm lengths for free-hand sectioning.	中文: 每片叶取最宽部位，切取 1 cm 长的叶段，用于徒手切片。
4	EN: Leaf sections were cleared using a 10% NaOH aqueous solution, soaking the leaf sections in it for two days.	中文: 叶段使用 10% NaOH 水溶液进行透明化处理，并在该溶液中浸泡两天。
5	EN: Once the leaves appeared transparent and softened, the NaOH solution was removed, and the leaf sections were rinsed three times with water.	中文: 待叶片呈透明并软化后，去除 NaOH 溶液，并用清水冲洗叶段三次。
6	EN: Subsequently, the leaves were dehydrated using an ethanol (EtOH) dilution series (30%, 50%, 70%, 100%), with each change taking no more than 5 minutes, followed by removing of the liquid.	中文: 随后，使用乙醇（EtOH）梯度对叶片进行脱水处理，浓度依次为 30%、50%、70% 和 100%。每次换液时间不超过 5 分钟，之后去除液体。
7	EN: After the 100% EtOH stage, the leaf samples were immersed in 1% safranin (1 g safranin/100 ml of 100% EtOH) for 10 minutes.	中文: 完成 100% EtOH 步骤后，将叶片样品浸入 1% 番红溶液中染色 10 分钟。该溶液的配制方法为每 100 ml 100% EtOH 中加入 1 g 番红。

8	EN: Then, the samples were gently rinsed with 100% EtOH and vacuumed off the liquid.	中文: 随后，用 100% EtOH 轻轻冲洗样品，并通过真空方式去除液体。
9	EN: Next, the samples were covered in 1% fast green (1 g fast green/100 ml of 100% EtOH) for a few seconds, followed by gentle rinsing with 100% EtOH and vacuuming off the liquid.	中文: 接着，将样品覆盖于 1% 固绿溶液中数秒。该溶液的配制方法为每 100 ml 100% EtOH 中加入 1 g 固绿。随后，用 100% EtOH 轻轻冲洗，并通过真空方式去除液体。
10	EN: The samples were then brought back to water through a reverse dilution series.	中文: 之后，样品通过反向稀释梯度重新转入水相。
11	EN: After the water stage, the leaves were mounted on slides using water for vein imaging.	中文: 进入水相阶段后，以水作为封片介质将叶片置于载玻片上，用于叶脉成像。
12	EN: The leaves were imaged under x10 magnification using a CCD camera (NIS-F1 Nikon, Tokyo, Japan) attached to a light microscope (Leica Microsystems AG, Germany).	中文: 叶片在 10 倍放大倍数下进行成像，使用连接于光学显微镜（Leica Microsystems AG，德国）的 CCD 相机（NIS-F1 Nikon，日本东京）拍摄。
13	EN: The images were analysed using ImageJ software (Al-Salman et al., 2023).	中文: 图像使用 ImageJ 软件进行分析，参考 Al-Salman et al. (2023)。
14	EN: The anatomical characteristics evaluated included vein density (VD), thickness of major vein and interveinal distance (IVD).	中文: 测定的解剖学特征包括叶脉密度（VD）、主脉厚度以及脉间距（IVD）。
15	EN: VD was calculated as the total length of longitudinal veins per unit leaf area (mm mm^{-2}), and IVD (μm) was defined as the average distance between consecutive minor longitudinal veins.	中文: VD 按单位叶面积内纵向叶脉总长度计算，单位为 mm mm^{-2} 。IVD 的单位为微米（ μm ），定义为相邻小纵向叶脉之间的平均距离。

Note: This version restructures the original methods paragraph into sentence aligned English Chinese pairs for easier reading and checking.